HEAT(q)

The heat (q) is regarded as an energy in transit across the boundary separating a system from its surrounding. Heat changes lead to temperature differences between system and surrounding. Heat is a path function.

Units of heat:

The SI unit of heat is joule (J). Heat quantities are generally measured in calories (cal). A calorie is defined as the quantity of heat required to raise the temperature of 1 gram of water by 1° C in the vicinity of 15° *C*.

Sign convention of heat:

The symbol of heat is *q*.

If heat flows into the system from the surrounding, energy of a system increases. Hence it is taken to be positive (+q).

If heat flows out of the system into the surrounding, energy of the system decreases. Hence, it is taken to be negative (–q).

Work(w)

Work is defined as the force (F) multiplied by the displacement(*x*).

$-w = F.\ x$ ----------(7.1)

The negative sign (–) is introduced to indicate that the work has been done by the system by spending a part of its internal energy.

The work,

(i) is a path function.

(ii) appears only at the boundary of the system.

(iii) appears during the change in the state of the system.

(iv) In thermodynamics, surroundings is so large that macroscopic changes to surroundings do not happen.

Units of work:

The SI unit of work is joule (J), which is defined as the work done by a force of one Newton through a displacement of one meter (J= Nm). We often use kilojoule (*k*J) for large quantities of work. 1 kJ = 1000 J.

Sign convention of work:

The symbol of work is '*w*'.

If work is done by the system, the energy of the system decreases, hence by convention, work is taken to be negative (– w).

If work is done on the system, the energy of the system increases, hence by convention, the work is taken to be positive (+w).

Pressure - volume work

In elementary thermodynamics the only type of work generally considered is the work done in expansion (or compression) of a gas. This is known as pressure-volume work, PV work or expansion work.

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